

EDUCATION

Princeton University

Ph.D., Electrical & Computer Engineering

2023 - 2028

- GPA 3.9/4.0; Ph.D. Fellowship in Natural Sciences and Engineering
- Research:** Computational Imaging & Optics, AI in Healthcare, Computer Vision, Neural Rendering

Sharif University of Technology

B.S., Electrical Engineering, Minor in Physics

2018 - 2022

- GPA 18.83/20; Top 10% of class; Academic Achievement Award (2019-2020); 12th/134K in National Entrance Exam
- Research:** Biomedical Signal Processing, AI in Healthcare, Wearable Sensors, Circuit Design, Embedded Systems

EXPERIENCE

Princeton University (Biomedical Imaging Group)

Princeton, NJ

Graduate Researcher (Advisor: Prof. Jason Fleischer; Collaborate with Prof. Felix Heide)

2024 - Present

- Developed unsupervised video-based resolution enhancement in fiber endoscopy with implicit neural representations
- Created a framework for manifold learning and alignment for personalized multi-lead ECG analysis.
- Implemented an algorithm for snapshot hyperspectral imaging with implicit neural representations

Waters Corporation (Industry PhD Partnership)

Remote, US

Computer Vision and Machine Learning Researcher

2024 - Present

- Design and implement a deep learning cell counting and viability detection system on brightfield and phase microscopy, robust to focal plane variations and staining strategies
- Driving Waters' next-generation imaging platform, recognized by senior leadership up to SVP level

Institute of Science and Technology Austria

Vienna, Austria

Research Intern (among 3% accepted applicants) (Advisor: Prof. Sandra Siegart)

Summer 2023

- Implemented 3D instance segmentation of microglia cells using classical computer vision techniques.

EPFL University (Integrated Neurotechnologies Lab)

Geneva, Switzerland

Research Intern (among 2% accepted applicants) (Advisor: Prof. Mahsa Shoaran)

Summer 2022

- Analyzed phase-amplitude cross-frequency coupling in EEG signals.

Sharif University of Technology (Biosen Group)

Undergraduate Researcher (Advisors: Profs. Mohammad Fakharzadeh & Daryoosh Vashayee)

2020 - 2022

- Designed a low-power wearable ECG patch for long-term cardiac monitoring with ML-based arrhythmia detection
- Researched energy harvesting materials and techniques for wearable medical devices.

TEACHING & MENTORSHIP

Princeton University

Teaching Assistant & Mentor

2024 - Present

- TA for Biomedical Imaging (ECE 452), Machine Learning & Pattern Recognition (ECE 435/535); Research Mentor

Sharif University of Technology

Teaching Assistant

2020 - 2022

- TA for Intro to ML, Signals & Systems, Computer Architecture, Applied Electronics in Bioengineering

PUBLICATIONS

- To be submitted:** Stain-Free Cell Viability Assessment with Deep Learning, A. R. Vazifeh, et al. *Frontiers in Optics*
- Under Review:** Manifold Learning for Personalized and Label-Free Detection of Cardiac Arrhythmias, A. R. Vazifeh, J. W. Fleischer. arXiv:2506.16494
- Manifold Alignment for Label-Free Cell Phenotyping in Multimodal Microscopy, A. R. Vazifeh, J. W. Fleischer. *Optica Biophotonics Congress 2026 — Microscopy, Histopathology and Analytics (Microscopy)*
- Seeing Through Fibers: Unsupervised Image Reconstruction in Fiber Bundle Imaging Systems, A. R. Vazifeh, et al. *Optics Express*, Feb 2026
- An open-source retrospective analysis of hypertrophic and dilated cardiomyopathy using machine learning and electrocardiogram data, A. Altintepe, et al. *Diagnostics*, Feb 2026
- Snapshot Hyperspectral Imaging via Compressive Sensing and Implicit Neural Representation, J. Boondicharern, A. R. Vazifeh, J. W. Fleischer. *Computational Optical Sensing and Imaging*, Aug 2025
- Advancing Personalized Healthcare: Progress in Energy Harvesting Materials of Self-Powered Wearable Devices, P. Bhatnagar, et al. *Progress in Materials Science* (Impact Factor: 40), Aug 2023
- Design and Implementation of an Ultralow-Power ECG Patch and Smart Cloud-Based Platform, B. Baraiejad, et al. *IEEE Trans. on Instrumentation and Measurement*, Apr 2022

SKILLS

Programming & ML - Python (PyTorch, TensorFlow, Sklearn, OpenCV, NumPy, SciPy), C++

Cloud & Data - AWS (S3, EC2), Large-scale data analysis, Distributed/Parallel computing

Hardware - Circuit Analysis & Board Design (Altium, HSPICE, Proteus), HDL (MIPS Assembly, Verilog)

Other - Git, LaTeX, HTML, CSS, Reviewer for IEEE Journal of Biomedical and Health Informatics (JBHI)